THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 13: **Statistics** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The class mark of a class is 25 and its width is 10. The class is: 1

(a) 20–30 **(b)** 25–35 **(c)** 15–25 **(d)** 20–25

2. The median of grouped data is given by: 1

(a) $l + \frac{\frac{n}{2} - cf}{f} \times h$ **(b)** $l + \frac{n - cf}{f} \times h$ **(c)** $l + \frac{cf - \frac{n}{2}}{f} \times h$ **(d)** $l - \frac{\frac{n}{2} - cf}{f} \times h$

3. If the mode is 24 and the mean is 60, then the median is: 1

(a) 48 **(b)** 42 **(c)** 36 **(d)** 52

4. In $\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$, d_i is: 1

(a) $x_i - a$ **(b)** $a - x_i$ **(c)** $\frac{x_i - a}{h}$ **(d)** $x_i + a$

5. A student states: “The mode is always greater than the mean.” The correct response is: 1

(a) Correct
(b) Incorrect — it can be greater, smaller or equal
(c) Correct for symmetric data
(d) Cannot be determined

6. For classes 0–10, 10–20, 20–30, 30–40, 40–50 with cumulative frequencies 4, 10, 18, 26, 30, the median class is: 1

(a) 10–20 **(b)** 20–30 **(c)** 30–40 **(d)** 40–50

7. Two students find the mean of the same data by the assumed mean method; A takes $a = 25$, B takes $a = 35$. Then: 1

(a) Only A is correct **(b)** Only B is correct
(c) Both get the same mean **(d)** Neither can get it

8. If $\Sigma f_i u_i = 20$, $\Sigma f_i = 50$, $a = 50$ and $h = 10$, the mean is: 1

(a) 54 **(b)** 52 **(c)** 46 **(d)** 50.4

9. The cumulative frequency of a class is: 1

(a) its own frequency
(b) the sum of frequencies up to and including that class
(c) half the total
(d) the class mark × frequency

10. The mean of 5 observations is 18. If one observation 22 is removed, the new mean is: 1

(a) 17 **(b)** 16 **(c)** 18 **(d)** 15

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of: 2

Class	0–10	10–20	20–30	30–40
Frequency	7	8	12	13

12. Find the mode of: 2

Class	0–20	20–40	40–60	60–80
Frequency	15	6	18	10

13. Find the median class and the modal class of: 2

Class	0–20	20–40	40–60	60–80	80–100
Frequency	6	9	20	15	10

14. A student says the mean of the data below is 30 (the average of the class marks). Justify or refute. **2**

Class	10–20	20–30	30–40	40–50
Frequency	2	4	6	8

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the mean daily expenditure on food by the step-deviation method: **3**

Expense (₹)	100–150	150–200	200–250	250–300	300–350
Households	4	5	12	2	2

16. Find the median life time of 400 lamps: **3**

Life (hrs)	1500–2000	2000–2500	2500–3000	3000–3500	3500–4000	4000–4500	4500–5000
Lamps	14	56	60	86	74	62	48

17. Find the mode of: **3**

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	10	35	52	61	38	29

OR

The mode of the following data is 36. Find the missing frequency x. **3**

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	8	10	x	16	12	6	7

18. Heights of 51 girls are given in "less than" form. Find the median height. **3**

Height less than (cm)	140	145	150	155	160	165
No. of girls	4	11	29	40	46	51

OR

The mean of the following distribution is 62.8 and $\Sigma f = 50$. Find p and q. **3**

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	5	p	10	q	7	8

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student obtains the median of the data below as 35. **5**

(i) Find the correct median.

(ii) Find the mean by the step-deviation method.

(iii) What error could the student have made?

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	10	18	12	5

OR

Find the median and mode of the wheat yield of 100 farms: **5**

Yield (qtl/ha)	50–55	55–60	60–65	65–70	70–75	75–80
Farms	2	8	12	24	38	16

20. **Case study:** Monthly electricity bills of 50 houses are given. **5**

(i) Write the modal class.

(ii) Find the mode.

(iii) Find the mean by the assumed mean method.

Bill (₹)	400–600	600–800	800–1000	1000–1200	1200–1400
Houses	7	10	15	12	6

OR

The median of the following data is 525 and $\Sigma f = 100$. Find x and y. **5**

Class	0–100	100–200	200–300	300–400	400–500	500–600	600–700	700–800	800–900	900–1000
Frequency	2	5	x	12	17	20	y	9	7	4

END OF QUESTION PAPER — **Total: 40 Marks**